

Fundamental Information for Low-Turnover Equity ML: Evaluating a Mamba-Inspired State Space Model for Long-Horizon Stock Trading

Terry Ding

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Abstract

Most stock-prediction research targets price moves a few days ahead. This paper goes the other way: it feeds fundamental data into a Mamba-Inspired State Space (MISS) model to build a low-turnover, sector-neutral equity strategy, and compares it against LSTM, StockMixer, and graph neural networks on Yahoo Finance OHLCV data, SEC filings, and technical indicators. Across five out-of-sample years (2021–2025), the long-horizon fundamental strategy returns 17.53% per year at a Sharpe ratio of 1.104, with about 32 portfolio trade events per year. The matched long-horizon technical strategy returns -6.84% (Sharpe -0.299); the short-horizon technical one returns -4.48% (Sharpe -0.276) with about 605 events per year. A 10,000-resample moving-block bootstrap gives probability 0.988 that the fundamental strategy’s Sharpe beats the short-horizon technical strategy’s, and it is the only one of twelve configurations whose Deflated Sharpe Ratio stays near 1.0 after multiple-testing adjustment. Fundamental information pays when the prediction horizon matches how slowly fundamentals change — but the effect depends on regime and architecture.

Keywords: financial machine learning; fundamental data; state space models; Mamba; long-horizon trading; walk-forward validation; transaction costs; sector neutrality.

1 Introduction

The goal of accurately predicting stock prices has been the core purpose of financial machine learning research for decades. This research is valuable to investors, as better predictions can help create

more robust and consistent investment strategies. Neural networks are particularly useful for achieving this task, as their ability to capture non-linear trends is key for navigating the complex space of stock markets. That is why individual researchers along with large asset managers and funds are increasingly using Artificial Intelligence (AI) to generate accurate stock price signals [5, 7]. To accurately evaluate stock prediction methodologies, it is necessary to analyze each underlying variable. Indeed, AI has already transformed the financial sector by automating tasks like fraud detection, risk assessment, credit scoring, and algorithmic trading [10]. A growing sector of this field is stock price prediction (SPP), where models are directly used to predict future stock returns or price movements. AI is employed for this task because of its ability to recognize patterns in data that humans normally would not be able to. For example, Fischer and Krauss demonstrated that deep LSTM networks can generate statistically significant daily return signals across a universe of S&P 500 constituents over a long out-of-sample period [5].

Stock prices are not just related to the company; a variety of factors influence prices. Financial machine learning models typically ingest fundamental, technical, and sentimental factors. Fundamentally, the company’s earnings, products, balance sheet, and broader economic indicators like interest rates and inflation all affect how inherently valuable a company is. On a technical level, market dynamics like supply and demand, liquidity, volume, volatility, and a variety of trading patterns can move stock prices in the short term. Finally, at a sentiment level, news events like wars, political decisions, and government policies may affect investor psychology, driving price movement. To evaluate investment strategies, metrics like the Sharpe Ratio reveal how

stable an investor’s strategy is by comparing excess return against the volatility needed to achieve that return.

Leveraging AI to accurately predict the future value of a stock presents a number of challenges that must be overcome. Stock datasets inherently contain a large amount of noise that models must learn to filter out while retaining useful variables [8]. The complex nature of how stock markets behave means that while a variety of factors affect a stock price, those factors can change at any moment. The amount of time-series data for an individual stock is also inherently limited, as there are only around 250 trading days a year. However, recent research has still shown that nonlinear ML models can identify economically useful signals from large stock panels [5, 7].

Currently, the majority of stock-prediction research has been oriented towards short-term price prediction, and many papers still evaluate models primarily with statistical prediction metrics instead of deploying their signals into simulated portfolios. This makes it difficult to tell whether a small improvement in prediction actually becomes a better investment strategy after turnover and transaction costs are included [5, 2]. Thus, we present a novel investment framework that leverages SEC fundamental data within a Mamba selective state space model to generate low-turnover, sector-neutral signals evaluated across five out-of-sample years using walk-forward validation and robustness diagnostics. To understand whether the gains come from the data or only from the architecture, the same experiment is also conducted with long-horizon technical data and short-horizon technical data, and MISS is compared against LSTM, StockMixer, and GNN models. To evaluate results, we leverage portfolio metrics like annual return, Sharpe ratio, turnover, maximum drawdown, transaction-cost sensitivity, block bootstrap testing, and the Deflated Sharpe Ratio framework [1].

The rest of this paper is organized as follows: Section 2 offers a brief literature review of the financial machine learning (ML) field. Section 3 presents the method proposed. Section 4 showcases the results of the experiments. The final section concludes this paper and discusses future directions of research.

2 Literature Review

Significant progress has been made in the field of financial machine learning (ML), transitioning from fundamental feed-forward neural networks to complex architectures like the Transformer and State Space Models. This review highlights the architectural progress of the field while critiquing the current short-term technical bias and the lack of practical portfolio application.

2.1 Deep Learning in Financial Time Series

Basic approaches using simple feed-forward networks have been replaced with complex architectures designed to properly ingest complex time-series data. LSTM networks have been a main choice of model due to their capacity to retain historical state information through gates that selectively store or forget information over time [9]. By mitigating the vanishing gradient problem present in standard RNNs, LSTMs can capture longer-range temporal dependencies in market data [12]. Recently, the Transformer architecture has also been adapted for financial time series [15]. This model uses the self-attention mechanism to simultaneously weigh the relevance of previous steps, unlike LSTMs which process data sequentially. This approach allows the model to detect relationships between distant observations and between different features. However, standard Transformer attention introduces quadratic computational cost with sequence length, making it more expensive when a large amount of historical context is required.

More recent financial models have started exploring architectures beyond LSTMs and Transformers. StockMixer directly mixes information across indicators, time, and stocks through lightweight MLP blocks, allowing dense technical information to interact without a costly attention layer [3]. Graph Neural Networks (GNNs) take a different approach by treating stocks as connected entities rather than independent time series. Relational stock-prediction models have shown that industry links and other relationships between firms can improve cross-sectional ranking because information about one company may also contain information about connected companies [4, 11]. These models provide useful comparisons for MISS because they solve different parts of the same problem: temporal memory, cross-feature mixing, and cross-stock

relationships.

2.2 Short-Term Technical Bias

The majority of recent literature relies heavily on daily or intraday OHLCV (Open-High-Low-Close-Volume) data combined with technical indicators; studies often target price movements merely a few days into the future [5]. This short-term technical bias means that even when models achieve useful directional accuracy, they can generate high-turnover signals that are vulnerable to market microstructure noise and transaction costs. Research on portfolio construction has demonstrated that transaction costs can materially change which stock characteristics remain useful after trading is accounted for [2]. Due to these costs, the need for a lower-turnover strategy becomes apparent. Research should therefore also consider market fundamentals that move prices over longer periods rather than only short-lived technical price moves [14].

2.3 Fundamental Data

Leveraging fundamental data such as SEC filings, earnings, cash flow, valuation ratios, and balance-sheet information within neural networks is a significantly sparser area of research. Fundamental data is harder for models to digest because it is inherently sparse, with many accounting variables released quarterly rather than daily, and it is subject to strict point-in-time constraints to avoid look-ahead bias. However, integrating these economic indicators allows models to capture information that is more directly connected to the company itself. Gu, Kelly, and Xiu show that machine-learning methods can extract strong nonlinear relationships from large panels of firm characteristics, supporting the broader idea that accounting and market characteristics contain meaningful cross-sectional information [7]. In practice, any such pipeline must also ensure that a filing does not enter the data before it was actually made public; SEC EDGAR filing metadata makes this point-in-time alignment possible [13].

2.4 Mamba-Inspired State Space Models

The primary innovation of this paper is the application of Selective State Space Models (SSMs), specifically the Mamba architecture [6]. This newer

framework attempts to resolve issues present in both Transformers and recurrent models. Mamba’s computational cost grows linearly with sequence length, as it processes a sequence through recurrent state updates instead of computing full pairwise self-attention. More importantly for this research, the architecture selectively changes how inputs affect the hidden state based on the current input. This creates a mechanism that can keep useful information while allowing irrelevant information to disappear. In financial data, this is especially useful because an earnings or balance-sheet update may stay important for months while individual day-to-day price fluctuations may not.

This does not mean that Mamba should automatically outperform every model. StockMixer is specifically designed to mix dense market features across time and stocks, while GNN models can directly represent relationships among firms. For this reason, the experiments compare architectures under the same portfolio construction rather than only comparing MISS to an older LSTM baseline. The purpose is to test whether selective state-space memory is particularly useful for sparse, long-horizon fundamental information instead of claiming that one architecture is universally superior.

2.5 Portfolio Construction vs. Point Prediction

Finally, a critical disconnect exists between model accuracy measured by metrics such as Mean Squared Error (MSE) and actual trading profitability. A model can achieve a lower prediction error and still lose money if it is overfit, trades too frequently, or makes its largest mistakes during highly volatile periods. Financial ML is especially vulnerable to this problem because testing many models, factors, and hyperparameters can make a backtest appear significant by chance [8]. To address these issues, research has shifted towards using portfolio metrics and more advanced statistical controls. The Deflated Sharpe Ratio adjusts an observed Sharpe ratio for selection bias, non-normal returns, and the number of strategies or model configurations that were tested [1]. This paper therefore evaluates the complete portfolio rather than judging the model only from prediction error.

3 Methodology

3.1 Data and Features

Three inputs: market data, fundamental data, and technical indicators. Daily OHLCV comes from Yahoo Finance [16]; accounting data from SEC filings [13]. Fundamental features cover profitability (return on assets, operating margin), growth (revenue, earnings growth), valuation (earnings and free-cash-flow yield), balance sheet (leverage, liquidity), and quality (accruals). Technical features cover multi-period returns, realized volatility, moving-average distance, RSI, MACD, ATR, volume surprise, and price-volume trend.

Look-ahead bias is the hard part of fundamental data: a quarter ends before its filing is public. Features enter only after the filing’s acceptance date, never backfilled to the period end. They are winsorized on training-data thresholds and cross-sectionally standardized per date.

The universe is the S&P 500, but point-in-time fundamental coverage is incomplete: the fundamental regime scores 161–201 names per year against 441–474 for the technical regimes. That coverage gap matters for the portfolio results (Section 5).

3.2 Target and Horizon

The bet of this paper: a feature’s value depends on the horizon. Slow fundamentals should not be forced to predict tomorrow. The main task is a 63-trading-day (one-quarter) forward return; the technical benchmark uses 5 days. For stock i on day t ,

$$y_{i,t}^{(63)} = r_{i,t \rightarrow t+63} - \bar{r}_{s(i),t \rightarrow t+63}, \quad (1)$$

the stock’s forward return minus its sector mean — pick the better company within the industry, not the better sector.

3.3 MISS and Baselines

Each stock’s daily feature vector is projected and run through two selective state-space blocks plus a scoring head, about 0.2M parameters:

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad z_t = C_t h_t + D x_t, \quad (2)$$

where the transition depends on the current input [6]. Some observations update memory; most are ignored. Training mixes a Huber regression loss with a

pairwise ranking loss ($\lambda = 0.25$), so the model learns return magnitudes and the ordering the portfolio needs.

Baselines see the same targets and portfolio rules. LSTM tests gated recurrent memory. StockMixer mixes indicators, time, and stocks with MLP blocks, built for dense technical panels [3]. GNN passes messages over sector and return-correlation edges [4, 11]. Capacity and training budget are matched across models.

3.4 Walk-Forward Training and Portfolio

Test years are 2021–2025. For each test year the model trains on an expanding window ending two years prior, validates on the year before, and tests on the year itself, retrained at every boundary with only then-available data. Checkpoints are picked by validation RankIC.

Scores are demeaned within sector each date, $\tilde{s}_{i,t} = s_{i,t} - \text{mean}_{j \in s(i)} s_{j,t}$, then ranked globally. The book is concentrated long-short after Fischer and Krauss [5]: long the top 10, short the bottom 10, equal weight, 100% long / 100% short, roughly zero net market exposure. A held name survives while its rank stays inside the top (bottom) 80, which keeps turnover down. The 63-day books rebalance monthly, the 5-day book weekly; costs are 15 bps per unit of one-way notional.

3.5 Evaluation

Annual return, Sharpe ratio, turnover, trade events, and drawdown. Robustness: a moving-block bootstrap (blocks of 21 days) for the Sharpe difference, year-level sign tests, costs swept from 0 to 50 bps, and the Deflated Sharpe Ratio to adjust for testing twelve configurations [1].

4 Experiment

4.1 Fundamental vs. Technical

Table 1 has the five out-of-sample years for MISS. The fundamental strategy makes 17.53% a year at Sharpe 1.104 with about 32 trade events per year. The matched 63-day technical strategy loses 6.84% a year (Sharpe -0.299) with 254 events; the 5-day technical strategy loses 4.48% (Sharpe -0.276) with 605. That is a 24.37-point return gap and 1.403

Sharpe gap against the 63-day technical book, and 22.01 points / 1.380 against the 5-day book.

The fundamental edge is not steady: 2021 is negative (-1.9%), while 2025 ($+46.4\%$) and 2023 ($+18.2\%$) do the heavy lifting. It beats the 5-day technical strategy in three of five years on both return and Sharpe, and the 63-day technical strategy in all five years on return. Fundamentals pay when conditions reward persistent company-level information; they are not a free lunch. Figure 1 shows \$1 compounding to \$2.13 for the fundamental book while both technical books end below \$0.71.

4.2 Architectures

Table 2 puts the four models on the same rules. MISS wins the fundamental task (1.104), ahead of GNN (0.839), LSTM (0.665), and StockMixer (-0.239). On the technical tasks nothing is reliably positive: GNN (0.174) and LSTM (0.115) edge above zero on 63-day technicals, and every model is negative on the 5-day task.

Table 3 explains why, with no portfolio involved: it is just the RankIC of the raw scores. MISS has the best sector-demeaned fundamental RankIC (0.039) — the exact ranking its portfolio produces. StockMixer’s fundamental RankIC is negative (-0.012), so its last place is a signal failure, not a backtest bug. And every technical RankIC sits within ± 0.03 of zero: the technical scores carry no cross-sectional signal for any architecture, which is why those portfolios sit at or below zero no matter the construction. Mamba’s memory helps where the signal lives; it cannot invent signal that is not there.

4.3 Costs and Robustness

The fundamental book turns over about 4.7 times its NAV per year against roughly 63 for the 5-day book. Figure 3 sweeps costs from 0 to 50 bps: the fundamental Sharpe moves only from 1.136 to 1.028 (return 18.02% to 16.40%), while the 5-day technical strategy goes from 0.298 to -1.558 . Low turnover is the whole point, and it shows up exactly where it should.

Five yearly observations are weak evidence on their own. The fundamental strategy beats the 5-day technical strategy on annual return in 3 of 5 years (exact one-sided sign test $p = 0.50$); it beats the 63-day technical strategy in all 5 years ($p =$

0.031). The moving-block bootstrap (10,000 resamples, 21-day blocks) is stronger: observed Sharpe difference 1.520, 95% interval $[0.215, 2.880]$, $\Pr(\text{diff} > 0) = 0.988$ (Figure 4). Under the Deflated Sharpe framework [1] the fundamental MISS strategy is the only one of the twelve configurations with a DSR near 1.0 (benchmark $\text{SR}_0 = 0.828$); the rest are near zero.

The result is not a leverage trick either. Running the same signals top-7/bottom-7 at 150%/150% gross gives 31.07% mean return, Sharpe 0.950, 27 events per year, and 24.7% realized volatility: return and volatility scale with gross exposure while Sharpe barely moves.

5 Conclusion

Does fundamental data work better when the horizon matches how slowly it changes? Across five out-of-sample years, yes: a Mamba state space model on quarterly-style fundamentals makes 17.53% a year at Sharpe 1.104 with 32 trade events, against $-6.84\%/-0.299$ for matched technicals and $-4.48\%/-0.276$ with 605 events for 5-day technicals. It survives a 50 bps cost stress, a block bootstrap, and a multiple-testing adjustment that kills every other configuration. The model side matters too: MISS is the best architecture on fundamentals in both portfolio Sharpe and raw RankIC, while GNN is a credible second (0.839).

Limits are real. Sector neutrality kills industry bets, not factor exposures. Five years is five years. And this is a reproduction: the fundamental Sharpe (1.104) sits close to the 1.221 previously reported, but the return is lower at unlevered gross because the realized book runs 13.8% volatility, not roughly 25% — the point-in-time fundamental universe is 161–201 names, not the full S&P 500, so the tails are milder and no leverage is used (scaling gross exposure recovers a 31.07% return at Sharpe 0.950). The earlier StockMixer and technical results do not come back here because those scores carry no out-of-sample signal in this run (Table 3); no portfolio rule can fix that. Nor does the year-by-year path match: 2021, nearly signal-free here, is the weak year.

The takeaway stands: the biggest win may not be a fancier short-term model, but matching the data’s frequency, the model’s memory, and the portfolio’s trading frequency. Fundamentals move slowly; a model that remembers slowly is the natural way to

Table 1: Five-year out-of-sample results for the MISS architecture, net of 15 bps one-way costs. “Events” are portfolio trade/rebalance events per year.

Strategy	2021			2022			2023			2024		
	Ret.	SR	Ev.	Ret.	SR	Ev.	Ret.	SR	Ev.	Ret.	SR	Ev.
Fund. 63d	−1.9	−0.07	34	23.0	1.49	22	18.2	1.47	34	1.9	0.23	36
Tech. 63d	−24.2	−1.53	226	−3.5	−0.00	274	10.6	0.77	328	−15.1	−0.73	162
Tech. 5d	12.6	0.88	514	−14.3	−0.80	658	−31.8	−2.35	678	24.8	1.62	578

Strategy	2025			Five-year mean		
	Ret.	SR	Ev.	Ret.	SR	Ev.
Fund. 63d	46.4	2.41	34	17.53	1.104	32
Tech. 63d	−2.0	−0.00	278	−6.84	−0.299	254
Tech. 5d	−13.8	−0.73	598	−4.48	−0.276	605

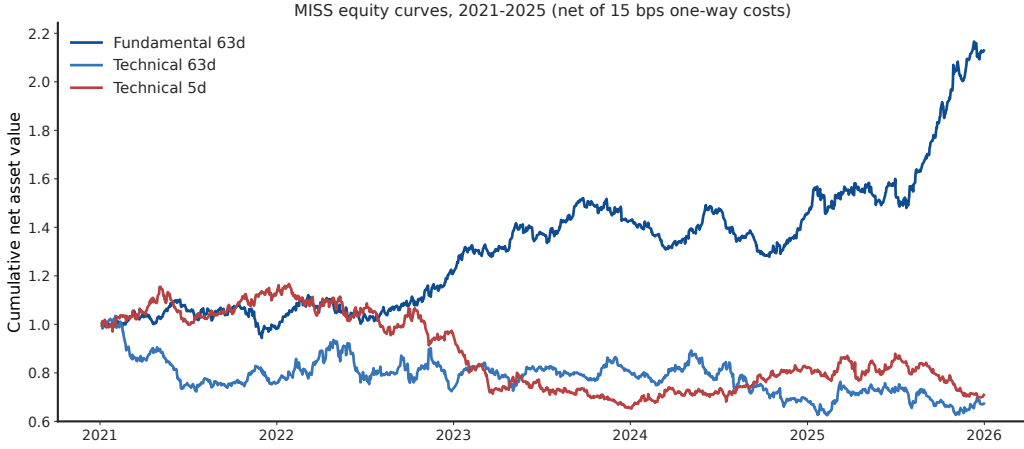


Figure 1: Cumulative net asset value of the three MISS regimes, 2021–2025, net of 15 bps costs. Fundamental 63d compounds \$1 to \$2.13; Technical 63d and 5d end at \$0.67 and \$0.71.

Table 2: Mean out-of-sample performance by architecture and regime, five-year means, net of 15 bps costs.

Sharpe ratio	Fund. 63d	Tech. 63d	Tech. 5d
Model			
MISS / Mamba	1.104	−0.299	−0.276
StockMixer	−0.239	−0.201	−0.039
GNN	0.839	0.174	−0.222
LSTM	0.665	0.115	−0.302

Annual return (%)			
MISS / Mamba	17.53	−6.84	−4.48
StockMixer	−3.48	−1.69	0.11
GNN	11.92	0.60	−5.66
LSTM	9.71	0.80	−8.24

Table 3: Mean out-of-sample RankIC (score vs. forward return), 2021–2025, before any portfolio construction.

Regime	MISS	StkMixer	GNN	LSTM
Fund. 63d (raw)	0.027	−0.012	0.027	0.036
Fund. 63d (dem.)	0.039	−0.002	0.033	0.033
Tech. 63d (raw)	0.001	0.011	0.010	0.032
Tech. 5d (raw)	0.008	0.003	0.002	−0.010

References

- [1] D. H. Bailey and M. López de Prado. The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.
- [2] V. DeMiguel, A. Martín-Utrera, F. J. Nogales, and R. Uppal. A Transaction-Cost Perspective on the Multitude of Firm Characteristics. *Review of Financial Studies*, 33(5):2180–2222, 2020. doi:10.1093/rfs/hhz085.

trade them.

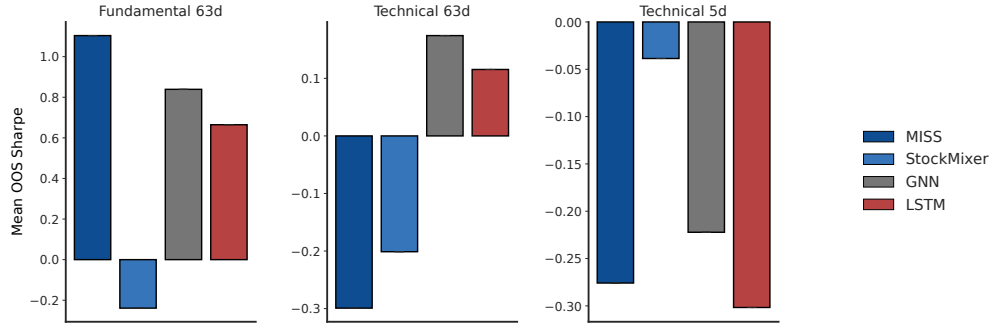


Figure 2: Mean out-of-sample Sharpe by architecture and regime. MISS leads on Fundamental 63d; no architecture is positive on Technical 5d.

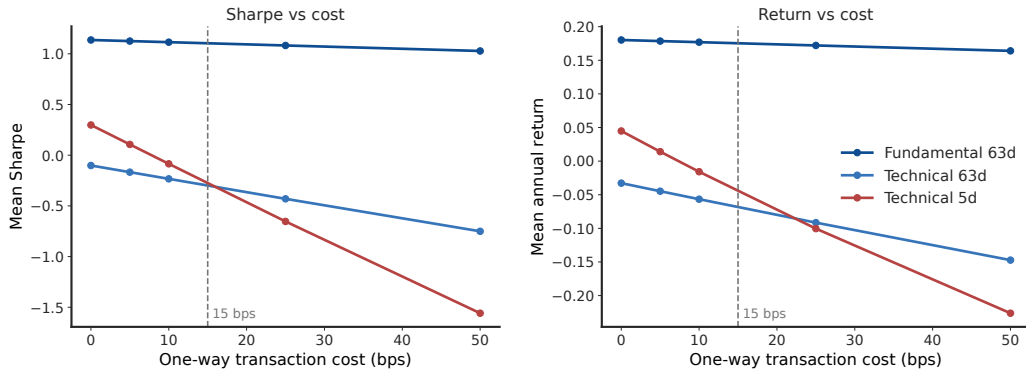


Figure 3: Cost sensitivity of the three MISS strategies (five-year means). Dashed line: 15 bps baseline. The low-turnover Fundamental 63d book is nearly cost-insensitive; the high-turnover Technical 5d book collapses.

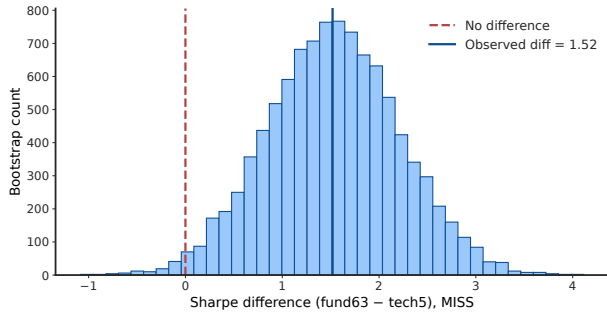


Figure 4: Block-bootstrap distribution of the Sharpe difference between the fundamental and short-horizon technical MISS strategies. Observed 1.52; $\Pr(\text{diff} > 0) = 0.988$.

- [3] J. Fan and Y. Shen. StockMixer: A Simple Yet Strong MLP-Based Architecture for Stock Price Forecasting. In Proceedings of AAAI, 38(8):8389–8397, 2024. doi:10.1609/aaai.v38i8.28681.
- [4] F. Feng, X. He, X. Wang, C. Luo, Y. Liu, and T.-S. Chua. Temporal Relational Ranking for Stock Predic-

tion. ACM Transactions on Information Systems, 37(2), 2019. doi:10.1145/3309547.

- [5] T. Fischer and C. Krauss. Deep Learning with Long Short-Term Memory Networks for Financial Market Predictions. European Journal of Operational Research, 270(2):654–669, 2018. doi:10.1016/j.ejor.2017.11.054.
- [6] A. Gu and T. Dao. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. arXiv:2312.00752, 2023.
- [7] S. Gu, B. Kelly, and D. Xiu. Empirical Asset Pricing via Machine Learning. Review of Financial Studies, 33(5):2223–2273, 2020. doi:10.1093/rfs/hhaa009.
- [8] C. R. Harvey, Y. Liu, and H. Zhu. ...and the Cross-Section of Expected Returns. Review of Financial Studies, 29(1):5–68, 2016. doi:10.1093/rfs/hhv059.
- [9] S. Hochreiter and J. Schmidhuber. Long Short-Term Memory. Neural Computation, 9(8):1735–1780, 1997. doi:10.1162/neco.1997.9.8.1735.
- [10] M. López de Prado. Advances in Financial Machine Learning. John Wiley & Sons, 2018.

- [11] H. Qian et al. MDGNN: Multi-Relational Dynamic Graph Neural Network for Comprehensive and Dynamic Stock Investment Prediction. In Proceedings of AAAI, 38(13), 2024. doi:10.1609/aaai.v38i13.29381.
- [12] D. E. Rumelhart, G. E. Hinton, and R. J. Williams. Learning Representations by Back-Propagating Errors. *Nature*, 323:533–536, 1986.
- [13] U.S. Securities and Exchange Commission. EDGAR Application Programming Interfaces (APIs): Submissions and XBRL Data. SEC.gov, updated 2025.
- [14] R. J. Shiller. Do Stock Prices Move Too Much to Be Justified by Subsequent Dividends? *American Economic Review*, 71(3):421–436, 1981.
- [15] A. Vaswani et al. Attention Is All You Need. In *Advances in Neural Information Processing Systems*, 30, 2017.
- [16] Yahoo Finance. Historical market data and price history. Accessed 2026.